Bhargav M

+91 7795036940 | bhargav.m23@iiits.in | LinkedIn | GitHub

Summary

B.Tech Artificial Intelligence and Data Science student graduating in 2027, with projects in Python data-processing pipelines, SQL databases, document extraction, and data validation. Builds traceable workflows with structured records, REST APIs, and automated checks. Seeking a **Data Engineering Intern** role at **Primenumbers Technologies**.

Work Experience

CYMAX Jan–May 2026

Unity Developer Intern

- Built Python services and a React playback client for VR streaming, integrating authentication, token-controlled access, and encrypted video delivery.
- Developed a Unity 360-degree video player for VR headsets with spherical rendering, headset/controller input, and playback controls.

Indian Institute of Information Technology, Sri City May–Jul 2026

XR and Simulation Intern

- Developed the Alipiri–Tirumala AR Navigation app as part of a collaborative project with **IIT Tirupati** and **Tirumala Tirupati Devasthanams (TTD)**.
- Processed GeoJSON route segments and landmark coordinates; combined GPS with visual-inertial localization for offline walking guidance.
- Developed a demo environment for ISRO's third launch pad using **Unreal Engine 5**.

Projects

Excel-to-App | Spreadsheet Data Processing and APIs GitHub

Python, SQL, FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Docker, pytest

- Built a pipeline to extract spreadsheet formulas, parse expressions, resolve dependencies, detect cycles, and evaluate calculations using a **66-function engine**.
- Stored immutable application versions and execution audit records; exposed REST APIs and validated results against LibreOffice, documenting **481/481 matching comparisons across 37 cases**.

ClaimBridge | Document Extraction and Reconciliation GitHub

Python, SQLite, Pydantic, PyMuPDF, BM25, Anthropic SDK, Streamlit

- Built a synthetic-claim review pipeline that extracts facts from PDFs with page-level provenance, reconciles them against JSON records, and retrieves policies by payer and effective date.
- Retained original evidence and reviewer corrections in SQLite; added citation-ID checks, audit records, approval, and tests for conflicting facts and missing documents.

StickerMe | Durable Jobs and Versioned Data GitHub

Python, FastAPI, SQLAlchemy, SQLite, React.js, TypeScript

- Built a background-job workflow for portrait-to-sticker generation, persisting jobs, engine attempts, output metadata, and saved revisions in SQLite.
- Implemented job leases, cancellation and recovery, version comparison and restore, and HTTP/server-sent-event integration for progress updates and downloads.

Alipiri AR Navigation | Geospatial Data Quality GitHub

C#, Unity, GeoJSON, JSON, GPS, A Pathfinding*

- Chained route segments by endpoint proximity, corrected segment direction, and validated landmark coordinates; flagged a **1.17 km source-data gap** as interpolated guidance.
- Used spatial indexing and reusable buffers for route queries and A* search; separated data access, navigation, and UI through interfaces and event-driven components.

Technical Skills

Programming and Analysis: Python, SQL, C#, Pandas, NumPy
Databases and APIs: PostgreSQL, SQLite, SQLAlchemy, FastAPI, REST APIs, JSON, Pydantic
Data Workflows: Spreadsheet parsing, PDF extraction, schema validation, reconciliation, audit trails
Engineering Tools: Git, GitHub Actions, Docker, pytest, data structures, dependency graphs

Education

Indian Institute of Information Technology, Sri City 2023–2027 (expected)

B.Tech in Artificial Intelligence and Data Science

Class XII: 92.3% – BGS PU College, Sidlaghatta | Class X: 97.6% – BGS Public School